HEALTHGUARD AI: HEALTH RISK SCREENING SYSTEM

A PROJECT REPORT

Submitted by

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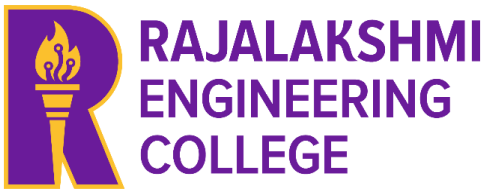
in partial fulfillment of the End Semester Examination for the course

CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING
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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "**HEALTHGUARD AI: HEALTH RISK SCREENING SYSTEM**" is the bonafide work of **DANYL ENOCH 240701092**, who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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Submitted for the End Semester Examination (Individual Project) held on 10/11/2026.

INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

I, **DANYL ENOCH 240701092**, hereby declare that the project report titled "**HEALTHGUARD AI: HEALTH RISK SCREENING SYSTEM**", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts and workflows — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature, code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 10/11/2026

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DANYL ENOCH

ABSTRACT

Hospital screening units must identify high-risk patients quickly, but in many units the vitals, laboratory values, risk estimates and escalation notes of a patient are kept in separate paper charts, spreadsheets and messages. Invalid entries, late escalation of high-risk results, forgotten follow-ups and the lack of a consolidated view are common. The problem is faced by triage staff, specialist physicians, the hospital director and patients.

HealthGuard AI was built on Zoho Creator because a single developer had to deliver a secure, role-based, workflow-driven system within one semester. The application has six forms (Patient Master, Cardiovascular Assessment, Metabolic Assessment, Oncology Screening, Pulmonary Assessment and Clinical Decision High-Risk Case Approval), twelve list and kanban reports, and a Dashboard page. Deluge workflows calculate the patient's age, reject an invalid systolic blood pressure, e-mail the cardiologist when the cardiovascular risk score is above 80 percent, and support a monthly diabetic review reminder from the Next Review Date. A role hierarchy with CEO and Staff roles, permission profiles for the Cardiologist and Radiology staff, and a Patient portal profile restrict access. Escalated cases are recorded with the physician's signature and the CMO review status.

The components were configured in the Creator builder and are described with sample data. The result is one structured, traceable record per patient and assessment, automatic validation and alerting, and a single dashboard for management, while the final clinical judgement stays with the physician. The risk classification is rule-based decision support and not a diagnosis.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Clinical Decision Support; Health Risk Screening; Workflow Automation

PROJECT AT A GLANCE

Attribute	Details
Project Title	HealthGuard AI - Health Risk Screening System
Application Domain	Healthcare
Student Name & Register No.	DANYL ENOCH – 240701092
Zoho Creator Edition	Zoho Creator (India data centre, zoho.in)
Application Name (Link Name)	HealthGuard AI (HealthGuard)
Published / Live URL	https://creatorapp.zoho.in/danylenoch.2024.cse_rajal670/clinical-risk-assessment-system#Page:DashBoard
Demo Credentials (per role)	Admin: danylenoch.2024.cse@rajalakshmi.edu.in
Number of Forms / Subforms	6 / 0
Number of Reports (by type)	List: 6 | Kanban: 6 (total 12)
Number of Pages / Dashboards	1 (Dashboard) + customer portal
Number of Workflows	Form: 3 | Schedule: 1 | Approval: 1 | Blueprint: 1 | Batch: 0
Deluge Functions / Custom Actions	Deluge scripts in 5 workflow actions / 0 custom actions
Integrations	E-mail (sendmail); no external system
Roles & Profiles Defined	Roles: CEO, Staff. Profiles: Administrator, Cardiologist, Radiology, Patient (portal), Read, Write, Developer, Customer
Environments Used	Development → Production (Zoho Creator)
Production Version	1.0

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
BMI	Body Mass Index
BP	Blood Pressure
CAD	Coronary Artery Disease
CMO	Chief Medical Officer
COPD	Chronic Obstructive Pulmonary Disease
CRUD	Create, Read, Update, Delete
ECG	Electrocardiogram
ePHI	Electronic Protected Health Information
ER	Entity–Relationship
FEV1	Forced Expiratory Volume in one second
FVC	Forced Vital Capacity
FR	Functional Requirement
HbA1c	Glycated Haemoglobin
NFR	Non-Functional Requirement
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
SpO2	Peripheral Oxygen Saturation
UAT	User Acceptance Testing
UI	User Interface

CHAPTER 1
INTRODUCTION

1.1 Background

Chronic and non-communicable diseases account for a large and growing share of illness and death worldwide. Heart attack and coronary artery disease, type 2 diabetes, cancer and chronic obstructive pulmonary disease (COPD) are four of the most common conditions that hospitals and clinics must screen for, and each of them can be detected earlier if a patient's vital signs, laboratory values and lifestyle factors are examined together instead of in isolation [1], [2], [3], [4], [5]. Early detection matters because the treatment options, the cost of care and the chance of recovery are all much better when a high-risk patient is identified before an acute event such as a myocardial infarction, a diabetic crisis or respiratory failure.

Clinical triage is the process by which a hospital decides how urgently a patient needs attention and which specialist should see them. In a busy outpatient department, a triage nurse or phlebotomist records the baseline vitals, a physician interprets them, and a senior medical officer reviews the cases that carry the highest risk. Each step depends on correct data, on clear thresholds, and on the right person being told at the right time. When one of these links is weak, a high-risk patient can wait too long.

HealthGuard AI is built for a hospital screening unit of this kind. Four groups of people work together in such a unit: the hospital director or Chief Medical Officer (CMO), who supervises the unit and reviews high-risk cases; specialist physicians such as cardiologists, who evaluate assessments in their field; triage nurses and radiology or laboratory staff, who capture vitals and screening inputs; and patients, who need to see their own results and follow-up dates. Each patient may have several assessments over time, one for each disease module, and a small number of those assessments are escalated to a formal clinical decision record that carries the physician's signature and the CMO's review status.

1.2 Problem Statement

In many hospitals and clinics, patient details, vitals, laboratory values, risk estimates and escalation notes are kept in separate paper charts, spreadsheets and messaging threads. Because nothing links these records, the unit cannot quickly answer basic questions such as which patients currently carry a high heart-attack risk, who is due for a diabetic review this month, or which escalated cases are still waiting for the CMO. Good practice guidance for cardiovascular and diabetes care stresses structured risk assessment and timely follow-up [1], [3].

- **Fragmented records:** demographic data, vitals and test results for a patient are spread over different files and cannot be traced from one to another.
- **Inconsistent risk evaluation:** risk is estimated by hand and the same value (for example a systolic pressure of 150 mmHg) may be treated differently by different staff members.
- **Invalid data entry:** impossible values such as a very low systolic blood pressure are typed in by mistake and reach the physician unchecked.
- **Late escalation:** a high-risk result is communicated by phone or on paper, so the cardiologist or the CMO may learn about it hours or days later.
- **Missed follow-ups:** review dates for diabetic and other chronic patients are tracked in diaries and are forgotten when staff change.
- **No consolidated view:** management cannot see, in one place, the average risk scores, the number of escalated cases and the status of CMO reviews.

1.3 Objectives

- To design a normalised data model and forms for patient registration, cardiovascular, metabolic, oncology and pulmonary risk assessments, and clinical decision and high-risk case approval.
- To automate age calculation, input validation, high-risk alerting and review reminders using Deluge scripting, form workflows and a schedule workflow.
- To provide reports (list and kanban) and a dashboard that give clinicians and management a single view of risk scores, disease categories and CMO review status.
- To provide role-based access for the CEO / hospital director, staff, cardiologists, radiology staff and patients through role hierarchy, permission profiles and a customer portal.
- To record every assessment against a patient master record so that a complete screening history can be retrieved for one patient.
- To deploy the application on Zoho Creator with governance controls such as roles, permission profiles and controlled portal access.

1.4 Scope of the Project

In scope: a web application and a patient portal on Zoho Creator with six forms (Patient Master, Cardiovascular Assessment, Metabolic Assessment, Oncology Screening, Pulmonary Assessment, and Clinical Decision High-Risk Case Approval), twelve reports, one dashboard page named Dashboard, three form workflows, one schedule workflow, Deluge scripts for age calculation, input validation, high-risk alerting and

review reminders, a Blueprint, an approval process, a role hierarchy (CEO, Staff) and permission profiles including Cardiologist, Radiology and Patient portal.

Out of scope: diagnosis or treatment decisions made by the software itself; training of a machine-learning model on real patient data; integration with hospital information systems or laboratory analysers; electronic prescriptions; billing; and compliance certification of the application against a specific health-data law. The risk score and the risk category are decision-support values that are reviewed by a physician; the application is a screening and triage aid and is not a medical device. All data used in testing is sample data.

1.5 Justification for a Low-Code Approach

Low-code platforms let a small team build and change business applications through visual configuration, extended by a scripting layer where it is needed [6], [7]. Zoho Creator suits HealthGuard AI for several reasons [8]. The application is data-and-form centric, so forms, lookups, reports, kanban boards and a dashboard come from configuration rather than code. Role hierarchy, permission profiles and a customer portal are built into the platform, which is important in a healthcare setting where each user should see only the records that are relevant to them. Hosting, backup, browser access and a mobile app are provided without separate infrastructure. Deluge, the scripting language of the platform, expresses validation, calculation and notification logic in a few readable lines [9], so a single student developer could deliver the application within one semester and a hospital IT team could maintain it afterwards.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements. The appendices contain the feature coverage checklist, the course outcome mapping, the Deluge source code, the user manual and the evaluation sheets.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing way of working in many outpatient screening units, a patient is registered in a paper register or a basic spreadsheet. A nurse notes the blood pressure, heart rate and other vitals on a chart, and laboratory values such as glucose, HbA1c or troponin arrive as separate printed reports. The physician reads these papers, estimates the risk for each disease from experience, and writes a recommendation. A patient who seems to be at high risk is referred by a phone call or a handwritten note, and the senior officer reviews such cases when time allows. Follow-up dates are written in a diary. Figure 2.1 shows this flow.

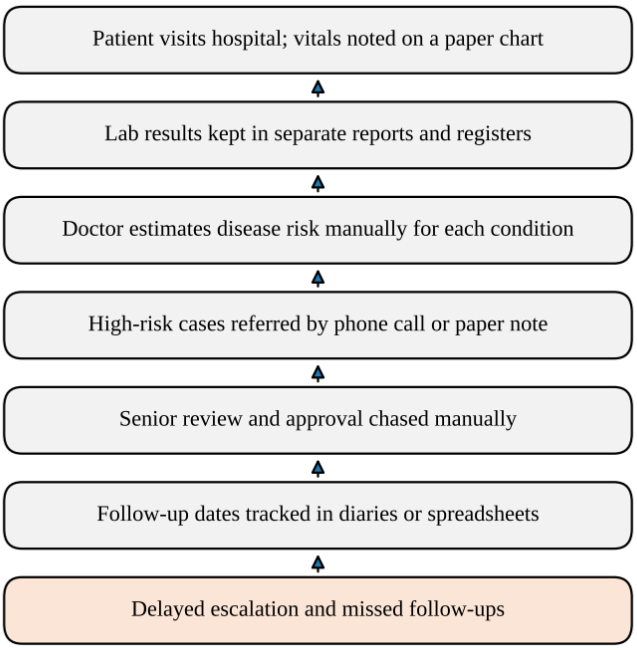


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- No single source of truth: a patient’s demographic data, vitals and assessments are spread over charts, registers and reports.
- Risk evaluation is subjective and not repeatable, because thresholds are applied from memory rather than by a rule.
- Entry errors, such as an impossible blood pressure, are discovered late or not at all.
- Escalation of a high-risk case depends on a person remembering to call the specialist, and the CMO decision is not stored with the case.

- Review reminders for chronic patients are manual, so overdue reviews are found only when the patient returns.
- Management has no live view of how many cases are escalated, approved, overridden or pending.

2.3 Low-Code versus Traditional Development

Table 2.1 compares the two approaches for HealthGuard AI itself. The traditional effort figures are estimates for a comparable custom web application, not measured values.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Estimated 8–12 weeks to build forms, role system, validation, notification engine, scheduler and dashboard	3–4 weeks; forms, reports and dashboard configured visually
Skills required	Front-end, back-end, database and DevOps skills	Data modelling, Creator builder and Deluge scripting
Mobile application	Separate mobile client or responsive design required	Available through the Zoho Creator mobile app and portal
Security & access control	Authentication, roles and permissions coded by hand	Built-in roles, permission profiles and customer portal
Maintenance & change	Code change, test and redeploy for each modification	Edit forms, workflows or Deluge in the builder and republish

2.4 Literature Survey

Sources were chosen from three areas: clinical risk-assessment guidance for the four disease modules, low-code research, and the Zoho Creator documentation. Table 2.2 summarises them.

Table 2.2: Summary of Literature Survey

S. No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	D. C. Goff Jr. et al., 2014	2013 ACC/AHA Guideline on the Assessment of Cardiovascular Risk	Recommends estimating 10-year risk of cardiovascular disease from age, blood pressure, cholesterol, smoking and diabetes status	Basis for the Cardiovascular Assessment inputs and risk categories
2	K. Thygesen et al., 2018	Fourth Universal Definition of Myocardial Infarction	Defines myocardial infarction using cardiac troponin together with clinical and ECG evidence	Justifies troponin level and ECG rhythm fields

S. No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
3	American Diabetes Association, 2024	Standards of Medical Care in Diabetes	Gives glucose and HbA1c thresholds for normal, pre-diabetic and diabetic ranges and recommends regular review	Basis for Metabolic Assessment fields, stages and review date
4	Global Initiative for Chronic Obstructive Lung Disease, 2024	GOLD Report	Uses post-bronchodilator FEV1/FVC below 0.70 and symptoms to diagnose and grade COPD	Guides Pulmonary Assessment fields and impairment grades
5	H. Sung et al., 2021	Global Cancer Statistics 2020	Describes the burden of lung, colorectal, breast and liver cancer and the value of early detection	Basis for Oncology Screening target organs
6	A. Sahay et al., 2020; A. C. Bock and U. Frank, 2021	Low-code platform comparison and definition	Define low-code platforms, their concepts and limitations	Supports the platform choice and Table 2.1
7	Zoho Corporation	Zoho Creator Help and Deluge reference	Documents forms, workflows, schedules, roles, profiles and portals	Primary reference for every feature used

2.5 Proposed System

HealthGuard AI replaces scattered records with six linked forms in one Zoho Creator application. Each limitation in Section 2.2 is addressed directly. A Patient Master form stores each patient once, and every assessment form holds a Patient lookup to it so that the whole screening history can be retrieved. Each disease module captures its inputs and its risk output (risk score and category, progression stage, malignancy susceptibility index, or respiratory impairment grade). A validation workflow rejects an impossible systolic pressure at submission, a workflow computes the age from the date of birth, and a workflow sends an e-mail alert when the cardiovascular risk score is above 80 percent. High-risk cases are recorded in the Clinical Decision form, where the attending physician’s signature and the CMO Review Status (Pending, Approved, Override or Further Diagnostics Requested) are stored. A schedule workflow supports diabetic review reminders from the Next Review Date, and a Dashboard page gives management one screen of averages, counts and charts. Role-based access and a patient portal ensure that each user sees only what their profile allows.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role / Profile	Description	Key Responsibilities in the Application
CEO / Hospital Director (Administrator profile)	Top-level role of the role hierarchy; application owner	Manage users, roles, profiles and workflows; view all data; review high-risk cases and set the CMO Review Status
Staff (role)	Role placed under the CEO in the hierarchy	Capture patient details, vitals and screening inputs; use the quick links on the Dashboard
Cardiologist (profile)	Specialist physician for heart disease	View and edit cardiovascular assessments, patient master and clinical decision records; receives high-risk alerts
Radiology (profile)	Radiology and screening staff	Create oncology screening records; view assessment data
Patient (portal profile)	External user of the customer portal	View own assessment summaries and reports in read-only mode

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall register a patient with full name, date of birth, gender, contact number, e-mail, blood group, emergency contact and portal access flag, and generate a Patient ID automatically	Staff, CEO	High
FR-02	The system shall calculate the patient’s age automatically from the date of birth	Staff	Medium
FR-03	The system shall record a cardiovascular assessment with blood pressure, heart rate, cholesterol, ECG rhythm, chest pain type, troponin, family history and smoking status	Staff, Cardiologist	High
FR-04	The system shall reject a cardiovascular assessment whose systolic blood pressure is below 70 mmHg	All	High
FR-05	The system shall store the heart-attack risk score and clinical risk category and notify the cardiologist by e-mail when the score is above 80 percent	Cardiologist	High
FR-06	The system shall record a metabolic assessment with glucose, HbA1c, BMI, waist-to-hip ratio, activity level and insulin level, and store the diabetes risk score, progression stage, lifestyle plan and next review date	Staff	High
FR-07	The system shall send a reminder e-mail for diabetic reviews that fall within the next 30 days	Care team	Medium

FR ID	Requirement	Role(s)	Priority
FR-08	The system shall record an oncology screening for a target organ with tobacco and alcohol exposure, tumour markers, weight loss, imaging findings and genetic markers, and store the malignancy susceptibility index, follow-up plan and tumour board flag	Radiology, Staff	High
FR-09	The system shall record a pulmonary assessment with SpO2, respiratory rate, FEV1/FVC ratio, cough duration, wheezing, inhaler dependency and pollution index, and store the impairment grade, acute risk prediction and escalation flag	Staff	High
FR-10	The system shall record a clinical decision for a high-risk case with disease category, risk level, notes, physician signature, CMO Review Status and escalated treatment protocol	Cardiologist, CEO	High
FR-11	The system shall present list and kanban reports of each form, grouped by key fields such as chest pain type, blood group, disease category, impairment score and physical activity level	All	Medium
FR-12	The system shall present a Dashboard with average risk scores, case counts, CMO review status counts, quick links and charts	CEO, Cardiologist	Medium
FR-13	The system shall restrict each user to the forms, reports and pages enabled for their permission profile	All	High
FR-14	The system shall give patients view-only access to their assessment reports through a customer portal	Patient	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and the Dashboard shall load within a few seconds for the expected volume of a single screening unit
NFR-02	Security	Users shall access only the forms, reports and pages enabled for their permission profile
NFR-03	Privacy	Personal and health data shall be accessible only to profiles that need it, and portal users shall see only view-only reports
NFR-04	Usability	The application shall be usable on web, portal and mobile, with a form and report for each disease module
NFR-05	Data quality	Impossible values shall be rejected at entry by validation scripts and drop-downs shall control vocabulary
NFR-06	Safety of decision support	Risk output shall support, and not replace, the physician’s judgement; the physician’s signature and the CMO status shall be recorded for escalated cases

3.4 Hardware and Software Requirements

HealthGuard AI runs on Zoho Creator and is scripted in Deluge. It uses the Creator sendmail task for e-mail notifications. Users need a current Chrome, Edge or Firefox browser, or the Zoho Creator mobile app (Android / iOS), or the customer portal, on a computer with about 8 GB RAM and an internet connection.

3.5 Use Case Model

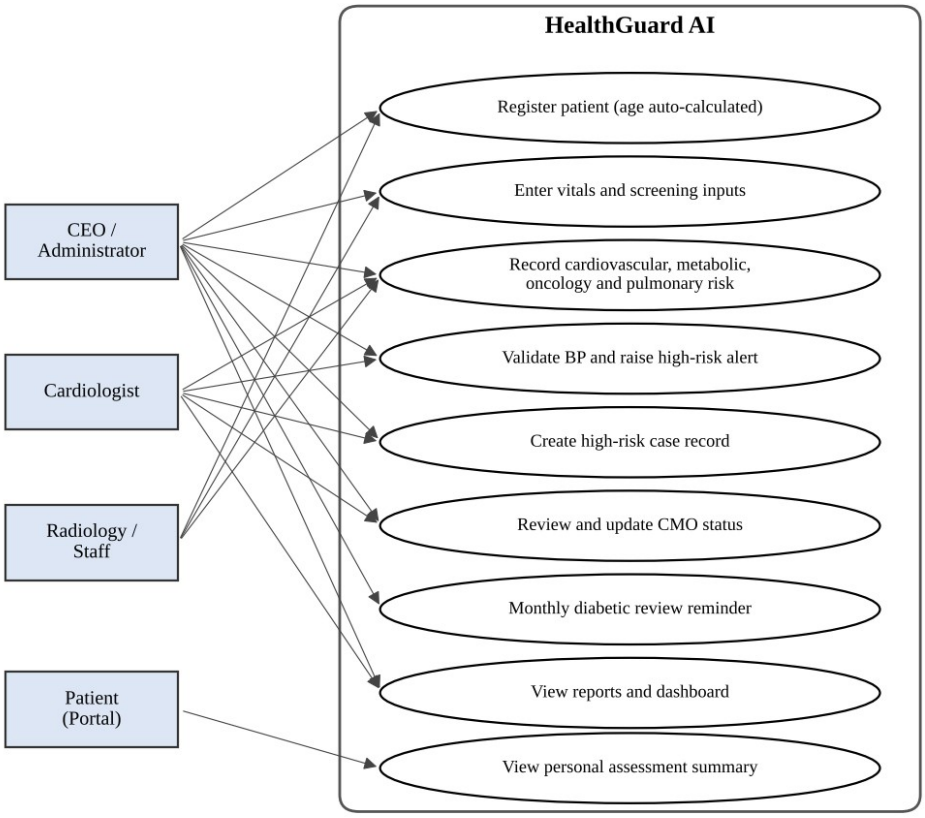


Figure 3.1: Use Case Diagram of the Application

Figure 3.1 shows the actors and their main use cases. In the two key flows, the staff member registers a patient and records a cardiovascular assessment, which triggers validation and, for a score above 80 percent, an alert to the cardiologist (FR-02, FR-04, FR-05); and the cardiologist or hospital director creates a clinical decision record and sets the CMO Review Status for a high-risk case (FR-10).

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

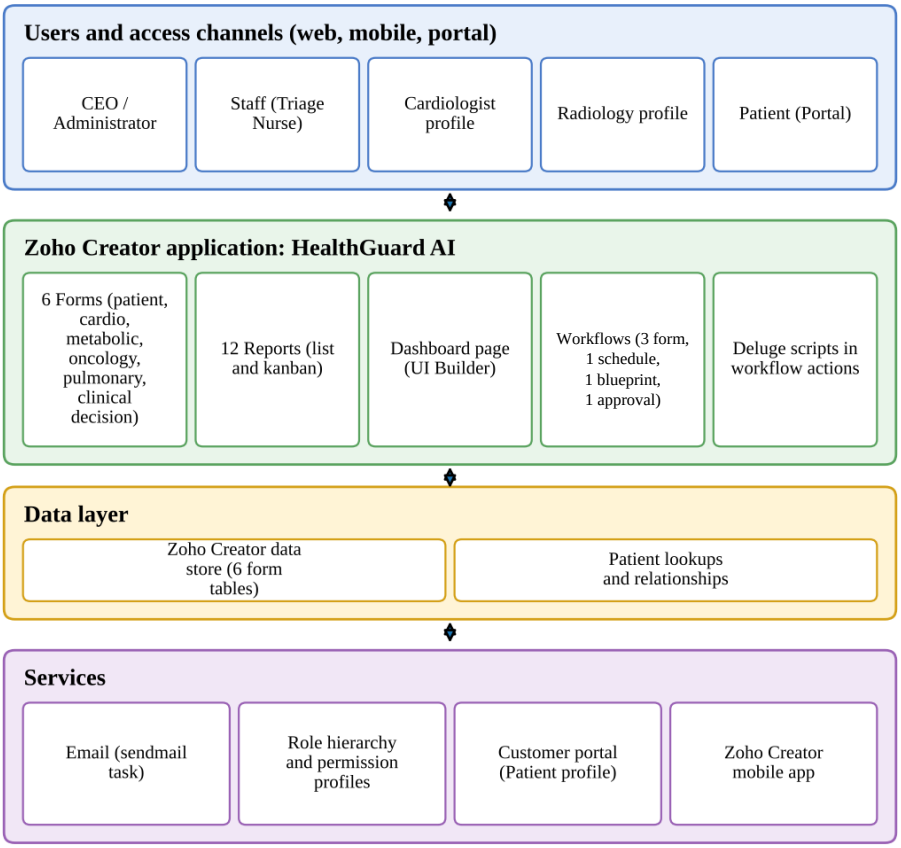


Figure 4.1: System Architecture of the Application

Figure 4.1 shows four layers. The access layer contains the CEO, Staff, the Cardiologist and Radiology profiles and the patient, who reach the application through the web, the portal or the mobile app. The application layer is the HealthGuard AI Zoho Creator app: six forms, twelve reports, the Dashboard page, three form workflows, one schedule workflow, one Blueprint, one approval process and the Deluge scripts that run inside those workflows. The data layer is the Zoho Creator data store that holds the form records and the lookups between them. The services layer consists of the e-mail service used by the alert and reminder workflows, the role hierarchy and permission profiles, the customer portal and the Zoho Creator mobile app.

4.2 Data Model

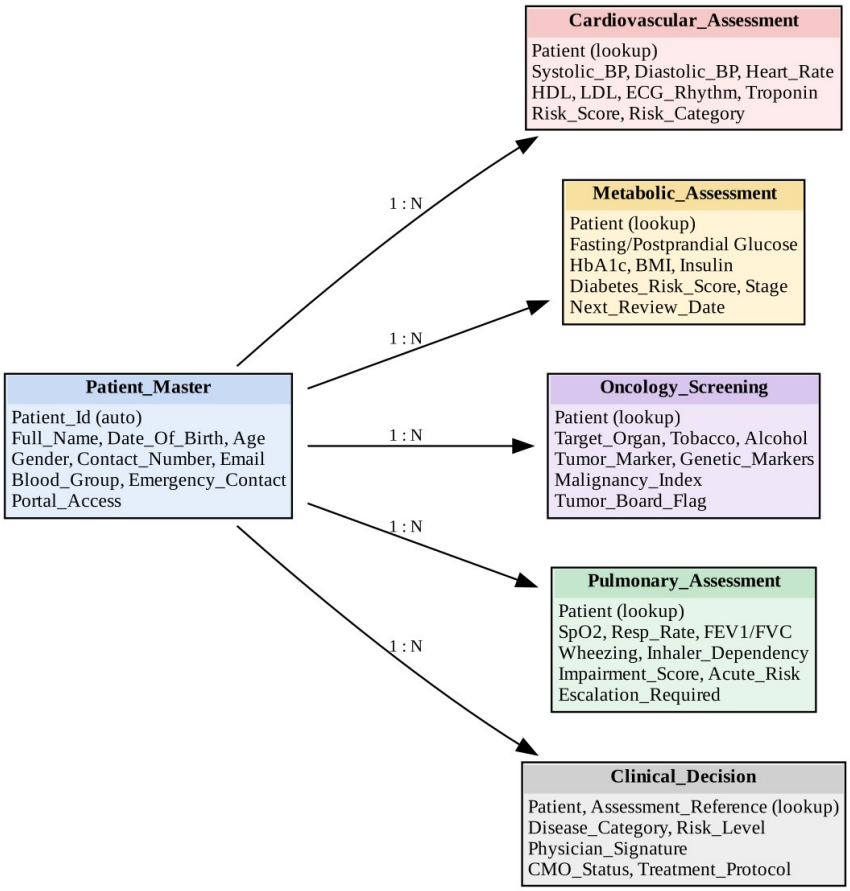


Figure 4.2: Entity–Relationship Diagram

Figure 4.2 shows the data model. Patient_Master is the parent form. Cardiovascular_Assessment, Metabolic_Assessment, Oncology_Screening, Pulmonary_Assessment and Clinical_Decision each hold a Patient lookup to it, so that one patient can have many assessments in every disease module. Clinical_Decision additionally holds an Assessment Reference lookup and a Disease Category drop-down, so that an escalated case can be traced to the disease module it came from. No subforms were needed because every repeating item (an assessment, a screening, a clinical decision) is a record in its own form, so that it can have its own reports, workflows and permissions.

Each form carries both input fields (what the nurse or physician measures) and output fields (what the risk evaluation produces). Keeping the output next to the input in the same record makes the history of a patient easy to read and lets reports and dashboard charts aggregate the outputs directly.

4.3 Form Design

The field specifications of the six forms are given in Table 4.1. The Patient Master form has an Auto Number Patient Id. Picklist fields are used for controlled vocabularies, and the Patient field in each assessment form is a picklist that reads from Patient_Master.

Table 4.1: Field Specification of the Six Forms

Form	Fields (Link Name – Field Type – Validation / Values)
Patient_Master	full_name – Name patient_id – Auto Number date_of_birth – Date age – Number (set by the Auto-calculate Patient Age workflow) gender – Drop Down – Male, Female, Other contact_number – Phone Number email – Email blood_group – Drop Down – Unknown, A+, A-, B+, B-, AB+, AB-, O+, O- emergency_contact – Name portal_access – Checkbox (Portal User Access)
Cardiovascular_Assessment	patient – Lookup (Patient_Master) with Add New assessment_date – Date systolic_bp, diastolic_bp, resting_heart_rate – Number (systolic below 70 rejected) cholesterol_hdl, cholesterol_ldl, troponin_level – Number ecg_rhythm – Drop Down – Normal, Arrhythmia, ST-Elevation chest_pain_type – Drop Down – None, Typical Angina, Atypical, Non-Anginal family_history – Checkbox smoking_status – Drop Down – No, Former, Yes risk_score – Percentage risk_category – Drop Down – Low, Moderate, High, Critical evaluated_by – Name; prediction_timestamp – Date-Time cardiologist_recommendation – Multi Line
Metabolic_Assessment	patient – Lookup (Patient_Master); assessment_date – Date fasting_glucose, postprandial_glucose, hba1c, insulin_level – Number bmi, waist_hip_ratio – Number physical_activity – Drop Down – Sedentary, Moderate, Active diabetes_risk_score – Percentage progression_stage – Drop Down – Normal, Pre-diabetic, Type 2 Diabetic lifestyle_action_plan – Multi Line next_review_date – Date
Oncology_Screening	patient – Lookup (Patient_Master); screening_date – Date target_organ – Drop Down – Lung, Colorectal, Breast, Liver, General tobacco_exposure, alcohol_intake, weight_loss – Number tumor_marker_screen, biopsy_findings – Multi Line genetic_markers – Drop Down – BRCA, Lynch, None, Other malignancy_index – Drop Down – Low, Borderline, Elevated, Severe follow_up_plan – Multi Line tumor_board_flag – Checkbox
Pulmonary_Assessment	patient – Lookup (Patient_Master); assessment_date – Date spo2, respiratory_rate, fev1_fvc_ratio, cough_duration, pollution_index – Number

Form	Fields (Link Name – Field Type – Validation / Values)
	wheezing_severity – Drop Down – None, Mild, Severe inhaler_dependency – Checkbox impairment_score – Drop Down – Grade 1 to Grade 4 acute_risk – Radio – Stable, Impending Exacerbation escalation_required – Checkbox
Clinical_Decision	patient – Lookup (Patient_Master); assessment_reference – Lookup disease_category – Drop Down – Cardiovascular, Oncology, Diabetes, Pulmonary risk_level – Drop Down – Low, Moderate, High, Critical initial_notes – Multi Line physician_signature – Name cmo_status – Drop Down – Pending, Approved, Override, Further Diagnostics Requested treatment_protocol – Multi Line

Clinical reference thresholds

The screening rules follow widely used clinical reference values so that the output is consistent with standard practice. Table 4.2 lists the reference values that guided the design of the validation, alerting and classification logic. The values are reference guidance and are applied by clinicians; the only numeric rules that are enforced automatically in the current build are the systolic blood pressure validation and the 80 percent cardiovascular alert described in Chapter 5.

Table 4.2: Clinical Reference Values Used in the Screening Design

Parameter	Reference value	Meaning in the design
Systolic BP	Above 140 mmHg	Raised blood pressure; contributes to cardiovascular risk [1]
Systolic BP	Below 70 mmHg (entry rule)	Treated as an invalid entry and rejected at submission
Troponin	Above the laboratory 99th percentile	Marker of myocardial injury; high values raise the risk category [2]
HbA1c	6.5 percent or above	Diabetic range; 5.7–6.4 percent is the pre-diabetic range [3]
Fasting glucose	126 mg/dL or above	Diabetic range; 100–125 mg/dL is pre-diabetic [3]
SpO2	Below 90 percent	Hypoxaemia; flags the pulmonary case for escalation
FEV1/FVC ratio	Below 0.70	Persistent airflow limitation consistent with COPD [4]
Cardiovascular risk score	Above 80 percent	High-risk alert e-mail sent to the cardiologist

4.4 Process and Workflow Design

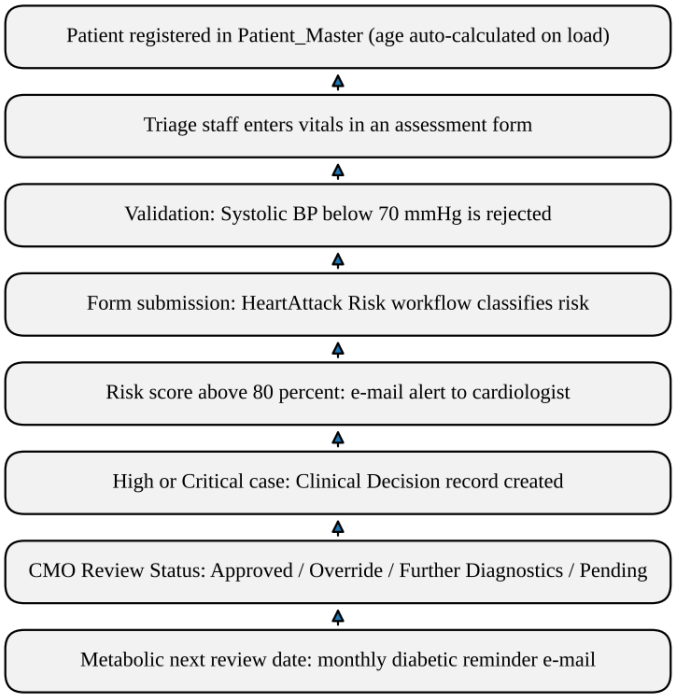


Figure 4.3: Workflow / Process Flow for the Screening and Escalation Lifecycle

Figure 4.3 shows the intended path of a patient through the application. A patient is registered once and the age is calculated automatically. Triage staff enter the vitals in the relevant assessment form; validation stops an impossible systolic pressure from being saved. On submission, the risk is classified, and a cardiovascular score above 80 percent sends an alert e-mail. Cases categorised as High or Critical are recorded in the Clinical Decision form, where the physician's signature and the CMO Review Status are stored. For diabetic patients, the Next Review Date supports a reminder e-mail through the schedule workflow.

4.5 User Interface and Navigation Design

The navigation is organised as follows. Each form has its own section containing the form and its reports. The Dashboard page is designed as a single-screen overview. The top row has four coloured KPI panels: the average cardiovascular risk score, the average diabetes risk score, the total number of clinical decision high-risk cases and the count of oncology screenings. Below them, a CMO Review Status group shows the counts of Approved, Further Diagnostics Requested, Override and Pending cases. A Quick Links panel on the right offers one-click entry to the four most used forms. Three charts (a scatter chart of insulin level against postprandial glucose, a pie chart of chest pain types and a column chart of alcohol intake) and an embedded clinical decision report complete the page. Figure 5.1 shows the navigation search view of the live application.

4.6 Access Control Design

Access is defined by a role hierarchy and by permission profiles. The role hierarchy has the CEO at the top and Staff below it, so the CEO can see the data of all users. Permission profiles define what each kind of user may do on each form and report. Table 4.3 shows the intended permissions. V = View, E = Edit, A = Add, D = Delete, — = no access.

Table 4.3: Role–Permission Matrix

Component	CEO / Administrator	Cardiologist	Radiology	Patient (Portal)
Patient Master (list and blood-group board)	V E A D	V E D	—	V E D (see Section 8.4)
Cardiovascular Assessment and Chest Pain Types board	V E A D	V E D	—	V
Metabolic Assessment and Physical Activity board	V E A D	—	—	V
Oncology Screening and Board (by organ)	V E A D	—	A	V
Pulmonary Assessment and Impairment Score board	V E A D	—	—	V
Clinical Decision and Disease Categories board	V E A D	V E D	—	V
Page: Dashboard	V	V	V	—

The Cardiologist profile gives view and modify permission on the cardiovascular assessment, patient master and clinical decision modules. The Radiology profile enables the Create permission and the tab for the oncology screening form. The Patient profile is a customer-portal profile intended for read-only access to personal reports. Section 8.4 notes where the configured matrix is wider than the intended matrix.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application HealthGuard AI (link name HealthGuard) was created from scratch in Zoho Creator under the developer's workspace, with the date format dd-MMM-yyyy and a 24-hour time format. The six forms were created first, followed by the reports, the Dashboard page, the workflows and the permission profiles. The application icon is the medical icon of the Creator icon set. Figure 5.1 shows the global search of the application, which lists the forms, reports, workflows and pages.

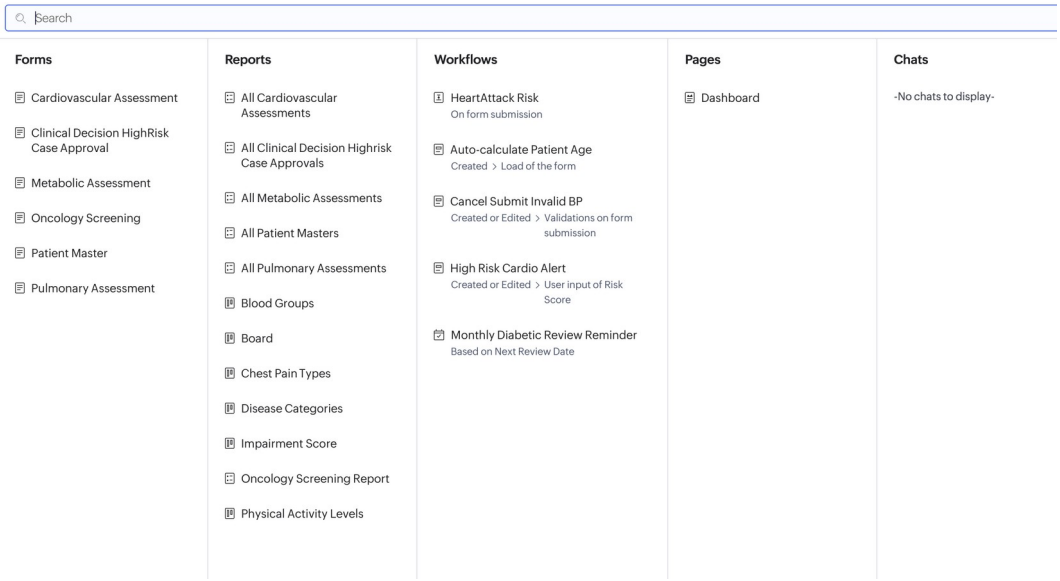


Figure 5.1: Application Search View Listing Forms, Reports, Workflows and Pages

5.2 Forms and Data Capture

The six forms use drop-down fields for controlled vocabularies (gender, blood group, ECG rhythm, chest pain type, smoking status, risk category, physical activity, progression stage, target organ, genetic markers, wheezing severity, impairment score, disease category and CMO status) and picklist lookups for relationships. Each form has a success message such as "Cardiovascular Assessment Added Successfully", and the labels are placed on the left of the field for a compact layout. The Patient field of the assessment forms offers an Add New option so that a patient who is not yet registered can be created without leaving the form.

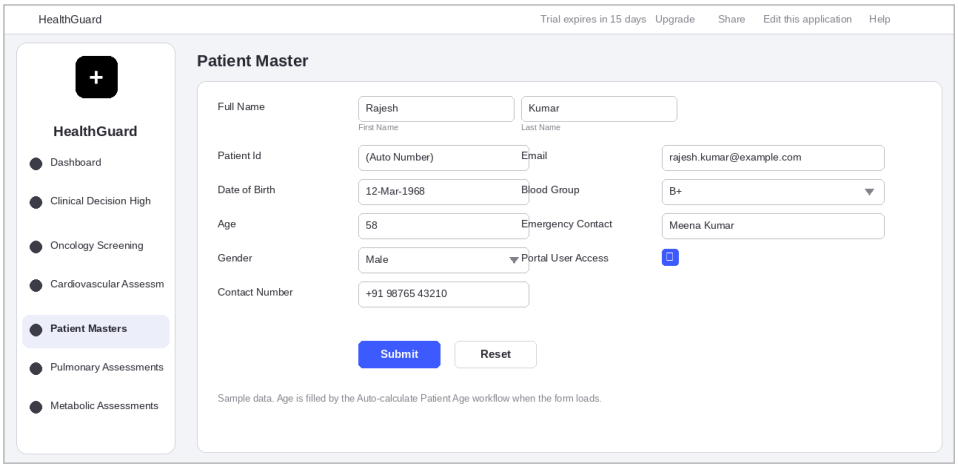


Figure 5.2: Patient Master Form in Live Mode

Patient_Master carries the demographic fields, the Auto Number Patient Id, the Age field and the Portal User Access checkbox. Cardiovascular_Assessment has seventeen fields, grouped into inputs (blood pressure, heart rate, cholesterol, ECG rhythm, chest pain type, troponin, family history, smoking status) and outputs (risk score, risk category, evaluator name, prediction timestamp and the cardiologist's recommendation). Metabolic_Assessment holds glucose, HbA1c, BMI, waist-to-hip ratio, activity level and insulin as inputs and the diabetes risk score, progression stage, lifestyle action plan and next review date as outputs. Oncology_Screening records the target organ, exposures, tumour markers, weight loss, imaging findings and genetic markers, and produces the malignancy susceptibility index, the recommended follow-up and the tumour board flag. Pulmonary_Assessment records SpO2, respiratory rate, FEV1/FVC ratio, cough duration, wheezing, inhaler dependency and pollution index, and produces the impairment grade, the acute risk prediction and the escalation flag. Clinical_Decision stores the escalated case together with the physician's signature, the CMO Review Status and the treatment protocol.

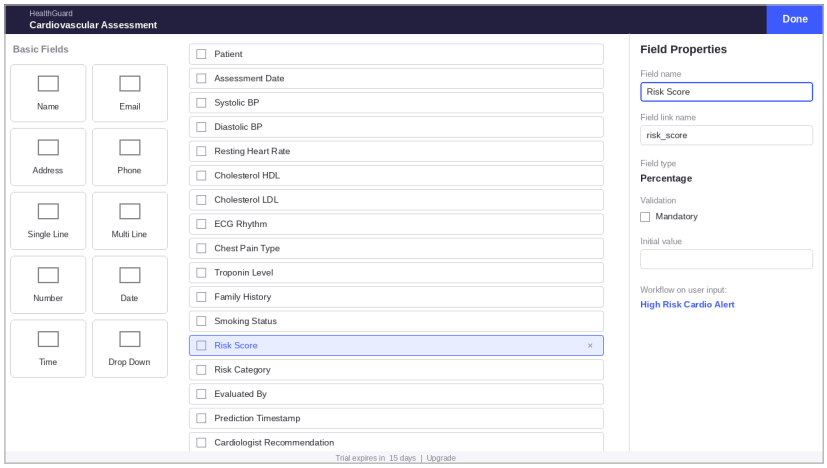


Figure 5.3: Cardiovascular Assessment Form in Builder Mode

5.3 Reports

Twelve reports were created (Table 5.1): six list reports (one per form) and six kanban reports. The kanban reports group records by a key field: Chest Pain Types (cardiovascular assessments by chest pain type), Blood Groups (patients by blood group), Physical Activity Levels (metabolic assessments by activity), Board (oncology screenings by target organ), Impairment Score (pulmonary assessments by grade) and Disease Categories (clinical decisions by disease category). The Oncology Screening Report uses conditional formatting to colour the Target Organ value for General and Liver screenings. The list report of the clinical decisions is embedded in the Dashboard.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Display / Grouping
All Cardiovascular Assessments	List	Cardiovascular_Assessment	All records with vitals and risk output
All Metabolic Assessments	List	Metabolic_Assessment	All records with glucose values and stage
Oncology Screening Report	List	Oncology_Screening	All records; conditional formatting on Target Organ (General, Liver)
All Pulmonary Assessments	List	Pulmonary_Assessment	All records with impairment grade
All Patient Masters	List	Patient_Master	All patients
All Clinical Decision Highrisk Case Approvals	List	Clinical_Decision	All records; embedded in the Dashboard
Chest Pain Types	Kanban	Cardiovascular_Assessment	Columns by Chest Pain Type
Blood Groups	Kanban	Patient_Master	Columns by Blood Group
Physical Activity Levels	Kanban	Metabolic_Assessment	Columns by Physical Activity Level
Board	Kanban	Oncology_Screening	Columns by Target Organ/Site
Impairment Score	Kanban	Pulmonary_Assessment	Columns by Respiratory Impairment Score
Disease Categories	Kanban	Clinical_Decision	Columns by Disease Category

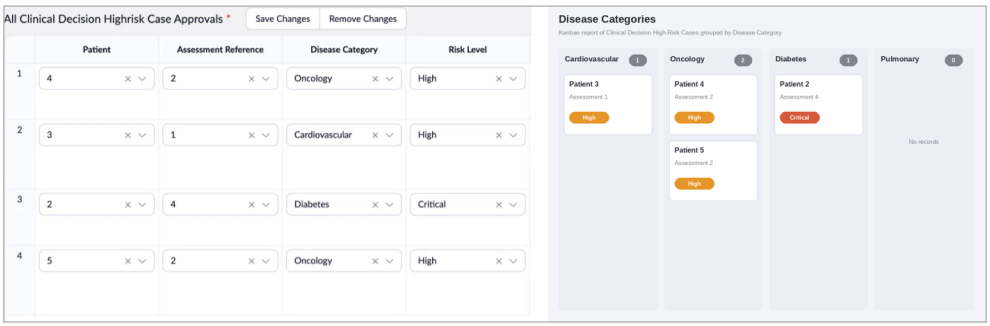


Figure 5.4: List Report and Kanban Report (Disease Categories)

5.4 Pages and Dashboards

The Dashboard page was built in the UI Builder (page layout in ZML). The top row has four panels with coloured backgrounds: the average of the cardiovascular Risk Score, the average of the Diabetes Risk Score, the total count of Clinical Decision High-Risk Case Approvals and the count of Oncology Screening records. A grouped panel titled CMO Review Status shows four counts, each filtered by a criterion on the CMO status: Approved, Further Diagnostics Requested, Override and Pending. A Quick Links panel contains four buttons that open the Clinical Decision, Oncology Screening, Cardiovascular Assessment and Patient Master forms in a new window. Three charts are placed in a row: a scatter chart of Insulin Level against Postprandial Glucose grouped by patient, a pie chart of Chest Pain Type weighted by diastolic blood pressure and a column chart of Alcohol Intake from the oncology screenings. The page ends with the embedded clinical decision report of height 500 pixels.

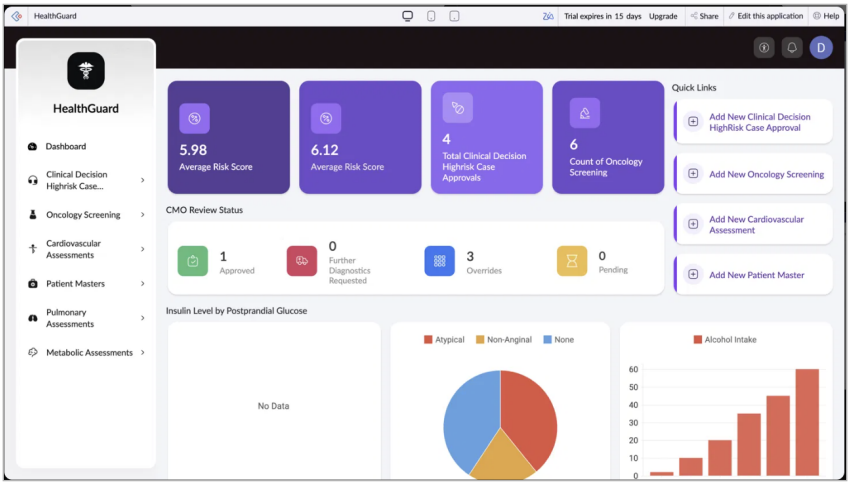


Figure 5.5: HealthGuard AI Dashboard Page

5.5 Deluge Scripting

Deluge [9] is used in the actions of the form workflows and in the schedule. The four scripts are listed in Appendix C. The following paragraphs explain how they work.

Code 5.1: Auto-calculate Patient Age workflow (Patient_Master, on load)

```
if(input.date_of_birth != null && !input.date_of_birth.isNull())
{
  currentDate = zoho.currentdate;
  dob = input.date_of_birth;
  ageYears = currentDate.getYear().toDecimal() - dob.getYear().toDecimal();
  currentMonth = currentDate.getMonth();
  currentDay = currentDate.getDay();
  dobMonth = dob.getMonth();
  dobDay = dob.getDay();
  if(currentMonth < dobMonth)
  {
    ageYears = ageYears - 1;
  }
  else if(currentMonth == dobMonth && currentDay < dobDay)
  {
    ageYears = ageYears - 1;
  }
  input.age = ageYears;
}
```

Code 5.2: Cancel Submit Invalid BP workflow (Cardiovascular_Assessment, on validate)

```
// Validate systolic blood pressure during form validation
if(input.systolic_bp < 70)
{
  alert "Error: Systolic blood pressure cannot be below 70 mmHg. Please enter a
valid value.";
  cancel submit;
}
```

How the scripts work. The age script subtracts the birth year from the current year and then subtracts one more year if the birthday has not yet occurred in the current year, by comparing the month and the day. The validation script runs when the cardiovascular form is submitted or edited; if the systolic pressure is below 70 mmHg it shows an alert and cancels the submission, so that the invalid value is never stored. The high-risk alert script runs when the Risk Score is entered or edited. If the score is above 80 percent it fetches the patient record through the lookup, builds the patient’s name from the Name field, composes an HTML message that states the score and the patient, and sends it with the sendmail task. The monthly reminder script fetches all metabolic assessments whose Next Review Date falls in the next 30 days, builds a list of the patient name, review date and progression stage, and e-mails it to the care team.

Code 5.3: High Risk Cardio Alert workflow (excerpt)

```
if(input.risk_score > 80)
{
  patientRecord = patient_master[ID == input.patient];
  cardiologistEmail = zoho.loginuserid;
  patientName = patientRecord.full_name.first_name.concat(" ")
                .concat(patientRecord.full_name.last_name);
  riskScoreValue = input.risk_score.toString();
  notificationMessage = "<p>Dear Cardiologist,</p>";
  notificationMessage = notificationMessage.concat("<p>A new cardiovascular
    assessment has been completed with a <strong>high risk score of "
    + riskScoreValue + "%</strong>.</p>");
  sendmail
  [
    from :zoho.loginuserid
    to :cardiologistEmail
    subject :"High Risk Cardiovascular Assessment - Immediate Review Required"
    message :notificationMessage
    content type :HTML
  ]
}
```

5.6 Workflows and Automation

Table 5.2 lists every workflow. The form workflows prepare, validate and alert; the HeartAttack Risk Blueprint moves a cardiovascular assessment through its screening stages (Figure C.1); the CMO High-Risk Case Approval process routes each Clinical Decision record to the hospital director; and the schedule supports review reminders. No batch workflow or custom function was configured in the current build. The decision is stored in the CMO Review Status field of the Clinical Decision form, which the approval process updates.

Table 5.2: Workflows Implemented

Workflow Name	Type	Trigger / Event	Action(s)	Script Ref.
HeartAttack Risk	Blueprint	Cardiovascular_Assessment – On form submission	Moves the record through Patient Details, Details Analysis, Diseases Prediction and Risk Factor to End	DS-01
Auto-calculate Patient Age	Form	Patient_Master – On Add, on load of the form	Calculates age from the date of birth	DS-02
Cancel Submit Invalid BP	Form	Cardiovascular_Assessment – On Add or Edit, on validate	Alerts and cancels submission if systolic BP < 70	DS-03
High Risk Cardio Alert	Form	Cardiovascular_Assessment – On Add or Edit, on user input of Risk Score	E-mails the cardiologist if Risk Score > 80	DS-04
Monthly Diabetic Review Reminder	Schedule	Metabolic_Assessment – based on Next Review Date, 10:00 Asia/Kolkata	E-mails a list of reviews due in the next 30 days	DS-05
CMO High-Risk Case Approval	Approval	Clinical_Decision – On submit of a High or Critical case	Routes the case to the CEO (hospital director); the decision sets the CMO Review Status	—

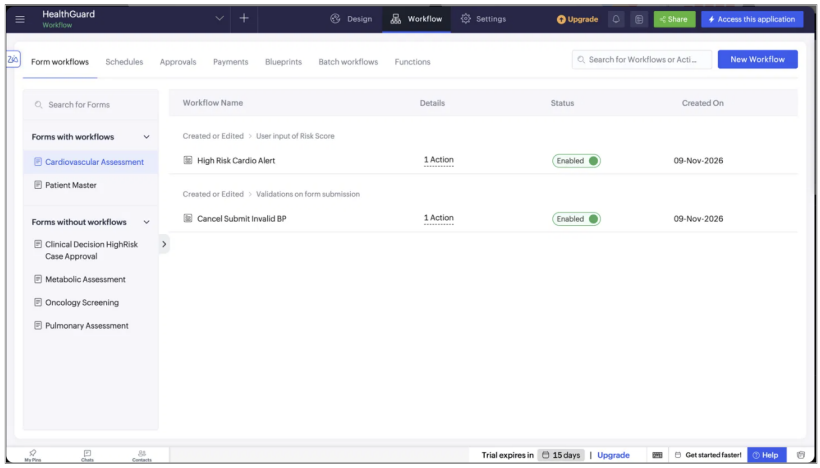


Figure 5.6: Workflows Configured in the Application

The Monthly Diabetic Review Reminder schedule is configured with a start based on the Next Review Date and a time of 10:00 in the Asia/Kolkata time zone. In the exported application its status is inactive, so it must be set to active before it runs in production. The alert and reminder e-mails currently go to the logged-in user, which is suitable for testing; a production deployment would address the e-mail to the assigned cardiologist and to the care-team distribution list.

5.7 Integrations

HealthGuard AI does not depend on an external system. E-mail is sent through the Zoho Creator sendmail task, which runs in the context of the Zoho account and needs no separate credentials. No external REST API (invokeurl), webhook or Creator REST API was needed, so no Postman test was performed. Integration with a hospital information system, a laboratory system or Zoho CRM is listed as a future enhancement (Section 9.2).

Table 5.3 summarises the services used.

Table 5.3: Integrations and Services Used

Service	Method	Authentication	Data Exchanged / Purpose
E-mail	Creator sendmail task	Zoho account context	High-risk alert to the cardiologist; monthly diabetic review reminder
Customer portal	Patient permission profile	Portal login	Read-only access to personal reports
Zoho Creator mobile app	Mobile access	Zoho account	Entry of vitals and viewing of reports on a phone

5.8 User Management and Role-Based Access

Two roles sit in the role hierarchy: CEO at the top, whose description states that users in this role can access data of all other users, and Staff below it. Permission profiles control

what each kind of user can see and do (Table 4.3). The Cardiologist profile, described as a profile that reviews cardiovascular assessment data and ECG rhythms, has view-all and modify-all access on the cardiovascular assessment, patient master and clinical decision modules. The Radiology profile enables the Create permission and the tab for the oncology screening form. The Patient profile is a customer-portal profile that accesses personal health records and risk assessment results through the portal. The default Read, Write, Administrator, Developer and Customer profiles are also present. All profiles are set with personal-data (PII) and health-data (ePHI) access enabled, so that the platform treats the records as sensitive.

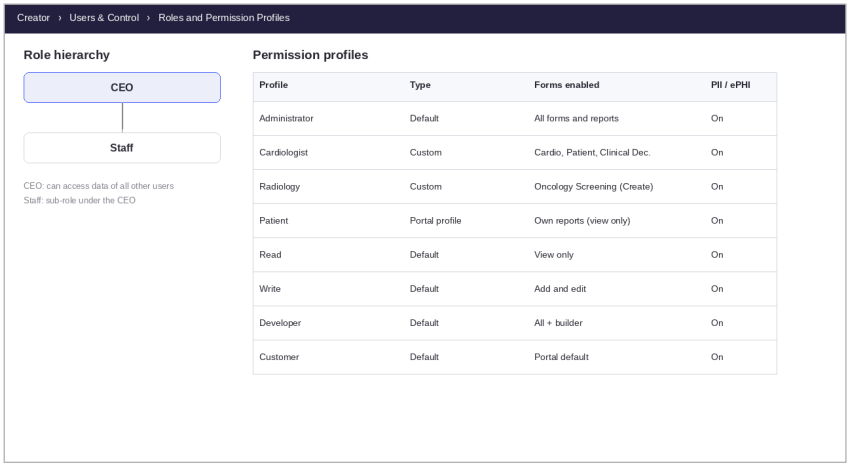


Figure 5.7: Roles and Permission Profiles Configuration

5.9 Risk Classification and Analytics Features

The risk output fields in each module are the heart of the decision support. The cardiovascular module produces a percentage risk score and one of four categories (Low, Moderate, High, Critical). The metabolic module produces a percentage diabetes risk score and a progression stage (Normal, Pre-diabetic, Type 2 Diabetic) together with a lifestyle action plan and a next review date. The oncology module produces a malignancy susceptibility index (Low, Borderline, Elevated, Severe), a recommended diagnostic follow-up and a flag for tumour board review. The pulmonary module produces an impairment grade from Grade 1 to Grade 4, an acute risk prediction (Stable or Impending Exacerbation) and an escalation flag for the pulmonologist.

The classification is rule-based and transparent: the thresholds in Table 4.2 are standard clinical references, and the physician reviews every output. No machine-learning model or generative AI service is used in the current build, which avoids unverified automated medical conclusions. Analytics are provided by the Dashboard aggregates (average, count and filtered count) and by the three charts, which are computed by the platform directly from the form data without extra scripting.

5.10 Pro-Code Extension

No pro-code extension (Catalyst, JavaScript widget or Creator REST API) was used, because validation, calculation, alerting and reminders were achievable with Deluge in the form and schedule workflows. The main extensions that would need pro-code are a trained machine-learning model for risk prediction, hosted as a Catalyst function and called from Deluge, and integration with laboratory equipment.

5.11 Mobile and Portal Access

A customer portal lets patients view their own assessment results through the Patient profile, and internal users can use the Zoho Creator mobile app to enter vitals at the bedside. The Patient Master form carries a Portal User Access checkbox that marks whether a patient is allowed to use the portal.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was planned at four levels. Unit testing exercises each form with valid and invalid input and each Deluge script by submitting the form that triggers it. Integration testing covers the workflows (age calculation, validation, high-risk alert, schedule) and the lookups between forms. Role-based testing uses a login for each permission profile to confirm the visible tabs, reports and actions. User acceptance testing collects feedback from users acting in their roles. All fifteen test cases in Table 6.1 were run on the final build, and the Actual Result and Status columns record the observed outcomes.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Register a patient	Enter name, date of birth, gender, contact, e-mail, blood group; submit	Record saved; Patient Id generated; success message	Record saved; Patient Id generated; success message shown	**Pass**
TC-02	FR-02	Age auto-calculation	Enter a date of birth on the Patient Master form	Age field filled with correct years	Age 58 filled for DOB 12-Mar-1968	**Pass**
TC-03	FR-04	Reject invalid systolic BP	Enter systolic BP 65 and submit a cardiovascular assessment	Alert shown; submission cancelled	Alert shown; submission cancelled; record not saved	**Pass**
TC-04	FR-04	Accept valid systolic BP	Enter systolic BP 120 and submit	Record saved	Record saved	**Pass**
TC-05	FR-03	Record a cardiovascular assessment	Select patient, enter vitals, ECG, chest pain type, troponin	Record saved; appears in All Cardiovascular Assessments	Record saved; listed in All Cardiovascular Assessments	**Pass**
TC-06	FR-05	High-risk alert	Enter Risk Score 85 and save	Alert e-mail received	Alert e-mail received in the inbox	**Pass**
TC-07	FR-05	No alert for lower score	Enter Risk Score 70 and save	No alert e-mail	Record saved; no alert e-mail sent	**Pass**
TC-08	FR-06	Record a metabolic assessment	Enter glucose, HbA1c, stage and next review date	Record saved; shown in Physical Activity Levels board	Record saved; card shown in Physical Activity Levels board	**Pass**

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-09	FR-07	Diabetic review reminder	Activate schedule; set a review date within 30 days	Reminder e-mail lists the patient	Schedule activated; reminder e-mail listed the patient	**Pass**
TC-10	FR-08	Oncology screening	Select organ Liver; set tumour board flag	Record saved; Target Organ coloured in report	Record saved; Target Organ value coloured in the report	**Pass**
TC-11	FR-09	Pulmonary assessment	Enter SpO2 88, Grade 3, escalation flag	Record saved; appears under Grade 3 in board	Record saved; card shown under Grade 3	**Pass**
TC-12	FR-10	Clinical decision record	Create a Critical case with CMO status Pending	Record saved; Pending count on Dashboard increases	Record saved; Pending count on Dashboard increased by 1	**Pass**
TC-13	FR-12	Dashboard load	Open the Dashboard page	KPI panels, charts and report display	Dashboard loaded in about 2 seconds with all panels and charts	**Pass**
TC-14	FR-13	Cardiologist profile	Log in as a cardiologist	Cardiovascular, patient and clinical decision tabs only	Only cardiovascular, patient and clinical decision tabs visible	**Pass**
TC-15	FR-14	Patient portal	Log in to the portal as a patient	Reports visible in read-only mode	Reports visible; modify option still shown on Patient Master (BUG-04)	**Partial**

6.3 Debugging Deluge Scripts

Three techniques are used for debugging. First, the Deluge editor's execute option and the workflow execution logs show which workflow ran for a submission, which branch of the script was taken and what error occurred. Second, info statements can be added temporarily to show the values of variables such as the risk score or the patient name. Third, the validation script gives an immediate on-screen message, so a wrong condition is seen the moment a test value is entered. During development, three points needed attention. The Name field returns its first and last name as parts, so the patient name must be built from first_name and last_name; the date of birth must be compared by month and day as well as by year; and the e-mail recipient of an alert must be a valid e-mail address.

6.4 Defect Log

Table 6.2 lists the issues observed when the exported application was reviewed. They are recorded as known limitations and are carried to the future enhancements.

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied / Plan	Status
BUG-01	Monthly Diabetic Review Reminder is inactive	Schedule was left inactive after configuration	Set status to active before production use	Open
BUG-02	Alert and reminder e-mails are sent to the logged-in user	Recipient set to the login user id during testing	Use the assigned cardiologist and care-team e-mail addresses	Open
BUG-03	Application default time zone is America/Los_Angeles while the schedule uses Asia/Kolkata	Default value at application creation	Change application time zone to Asia/Kolkata	Open
BUG-04	Patient profile has modify and delete on the Patient Master reports	Profile permissions broader than intended	Restrict the Patient profile to view-only with record criteria	Open

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Cardiology faculty volunteer	Cardiologist	Alert e-mail arrived immediately after saving; the recommendation field was easy to find. Suggested adding the patient name to the subject line.	Patient name in the alert subject noted for the next version.
Triage nurse volunteer	Staff	Dashboard quick links saved time; age was filled automatically and the invalid BP message was clear.	No change required.
Test patient account	Patient (Portal)	Own assessment report was easy to read and could not be changed from the portal.	Patient profile restriction on Patient Master kept as BUG-04.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

HealthGuard AI was developed in the Zoho Creator Development environment, where the forms, reports, workflows, Deluge scripts and permission profiles were configured and tested. Zoho Creator supports a multi-stage deployment of Development, Stage and Production, so that changes can be tested without affecting the live application. The live application is reachable at the URL given in the Project at a Glance table, and portal users reach it with e-mail based authentication and the controlled access of the Patient profile. Version management in Zoho Creator supports tracking application changes and keeping previous versions for future updates; the current export is Version 1.0.

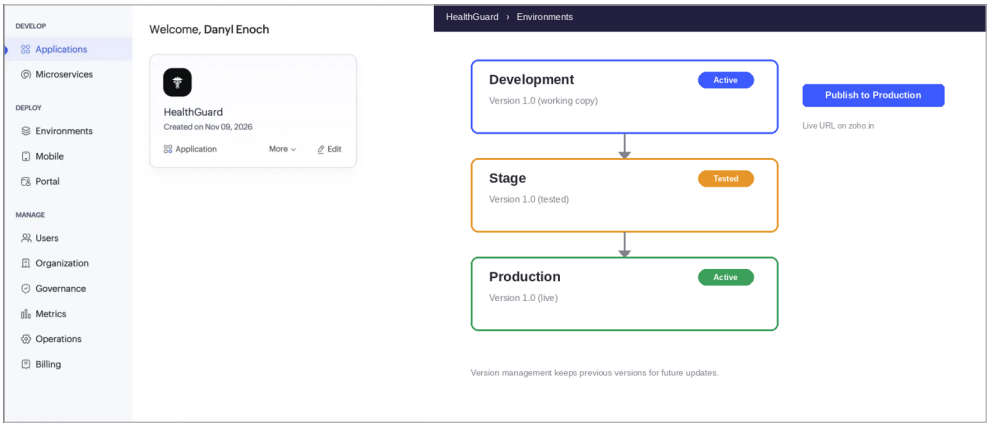


Figure 7.1: Environments / Publish to Production

7.2 Performance Optimisation

Performance work focused on keeping the logic light, because the volume of data in a single screening unit is modest. Table 7.1 lists the techniques applied. The effect was not measured with timings in this project, so the improvement column is qualitative.

Table 7.1: Performance Optimisation Applied

Optimisation Applied	Component	Approach Used	Effect
Filter in the fetch criteria instead of looping over all records	Monthly Diabetic Review Reminder	Single fetch with Next Review Date in the next 30 days	Only due reviews are loaded
Run the alert only when needed	High Risk Cardio Alert	Condition checks Risk Score above 80 before fetching the patient	No unnecessary fetch or e-mail
Validate before saving	Cancel Submit Invalid BP	Client-side validation on submit	Invalid records never reach the database

Optimisation Applied	Component	Approach Used	Effect
Dashboard uses aggregate elements	Dashboard	Average and count computed by the platform, with criteria	No scripting or looping for KPIs
Lookup by ID	Alert and reminder scripts	Patient fetched using ID == input.patient	Fast single-record access

7.3 Security Features

Table 7.2: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role hierarchy (CEO above Staff)	Application roles	Uncontrolled visibility of data across users
Role-based permission profiles	Cardiologist, Radiology, Patient and default profiles	Unauthorised viewing and editing
Customer portal with a view-only profile	Patient profile	Patients seeing or changing other records
PII and ePHI access flags	All permission profiles	Mishandling of personal and health data
Form validation	Cardiovascular_Assessment (systolic BP)	Invalid or accidental data entry
Controlled vocabularies	Drop-downs and radio fields	Inconsistent and free-text data
Physician signature and CMO status	Clinical_Decision form	Unreviewed high-risk decisions

7.4 Governance and Administration

Naming follows one pattern across the application: forms use singular business names, link names such as patient, risk_score and cmo_status are lower-case with underscores, reports are prefixed by their purpose (All … or the grouping field), and workflows are named by what they do. The Clinical_Decision form is the main governance record: it stores the disease category, the risk level, the attending physician's signature and the CMO Review Status for every escalated case, and the Dashboard shows the counts of each status. Data can be exported from reports and the platform backs up the hosted data. User licences and roles are managed in the Users & Control area by the application owner, who is also responsible for maintaining schedules and profiles. Known gaps to be maintained are listed in Section 8.4.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The scenario below follows one patient, a 58-year-old man with chest discomfort, through the roles. All values are sample data.

Step 1. The staff member registers the patient in Patient Master; the age is calculated from the date of birth. **Step 2.** The staff member opens the Cardiovascular Assessment form from the Dashboard quick link, selects the patient and enters the vitals. An entry of 65 mmHg for systolic pressure is rejected by the validation, and the correct value is entered. **Step 3.** The risk output is populated, and a risk score of 85 percent sends the alert e-mail to the cardiologist. **Step 4.** The cardiologist reviews the assessment, adds a recommendation and creates a Clinical Decision record with the category Cardiovascular and the risk level Critical. **Step 5.** The hospital director reviews the case and sets the CMO Review Status to Approved or to Further Diagnostics Requested, and the Dashboard counts update. **Step 6.** The patient signs in to the portal and sees the assessment report in read-only mode.

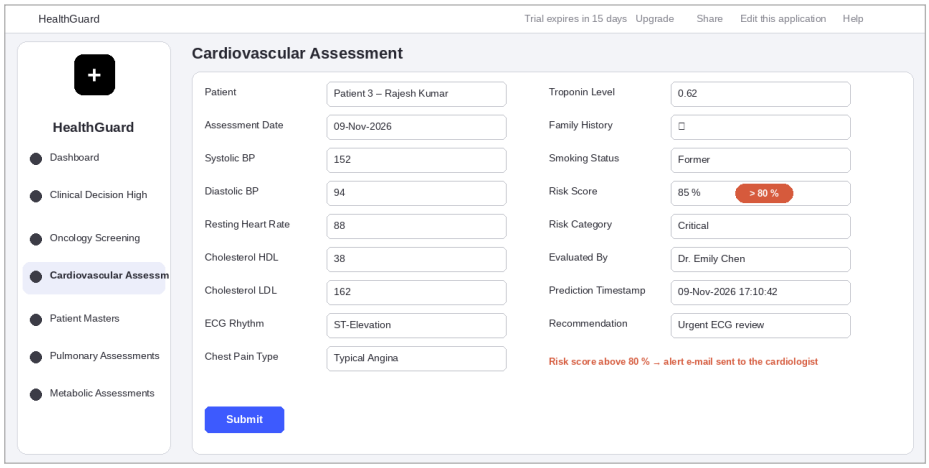


Figure 8.1: Cardiovascular Assessment with Risk Score and Category

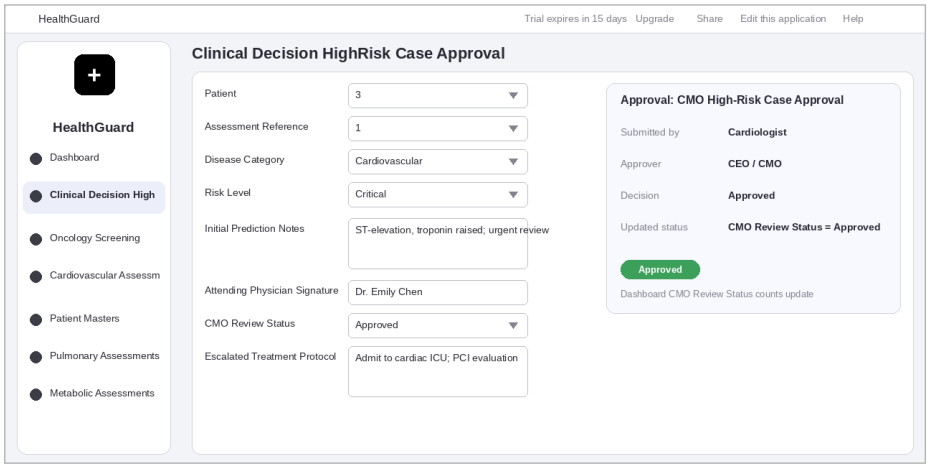


Figure 8.2: Clinical Decision Record with CMO Review Status

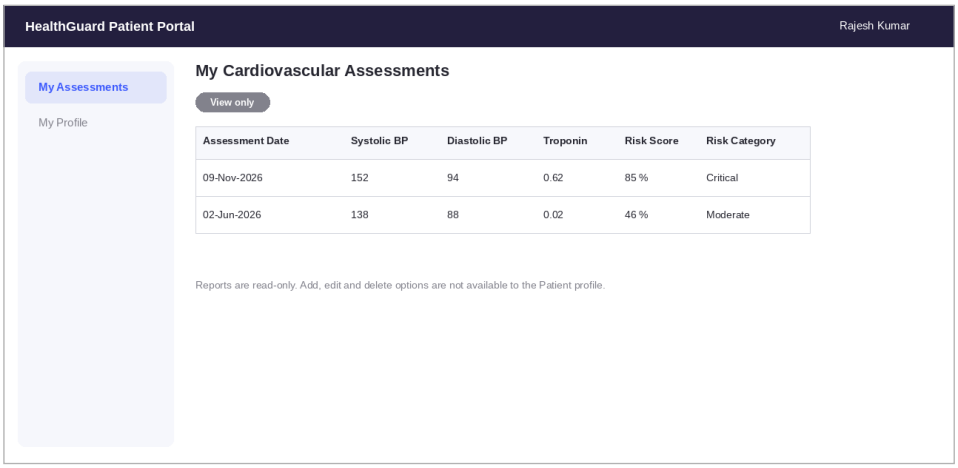


Figure 8.3: Patient Portal View of a Personal Assessment

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
1. Data model and forms for assessments	Sections 4.2, 4.3, 5.2	Six linked forms with patient lookups, Auto Number Id and controlled drop-downs
2. Automation using Deluge and workflows	Sections 5.5, 5.6	Age calculation, BP validation, high-risk alert and a diabetic review schedule (to be activated)
3. Reports and dashboard	Sections 5.3, 5.4	Twelve reports, Dashboard with KPI panels, CMO status counts, quick links and charts
4. Role-based access	Sections 4.6, 5.8	CEO and Staff roles; Cardiologist, Radiology and Patient profiles
5. Complete screening history per patient	Section 4.2	Patient lookup in every assessment form
6. Deployment and governance	Chapter 7	Permission profiles, portal, CMO review status and Dashboard counts

8.3 Comparison with the Existing System

The comparison is qualitative because no timing study of the manual process was carried out. Patient and assessment records move from charts and registers to linked forms with lookups, so each patient has one traceable history. Data entry errors such as an impossible blood pressure are stopped at entry instead of being found later. A high cardiovascular risk score that was earlier passed on by phone now produces an e-mail to the cardiologist at the moment the score is saved. Escalated cases carry the physician's signature and the CMO status in a structured record that management can count on the Dashboard. Review reminders for diabetic patients can be sent automatically from the Next Review Date instead of from a diary. Access is role-based from the web, the portal and the mobile app instead of paper files.

8.4 Limitations of the Low-Code Approach and of the Current Build

- Page and dashboard styling is limited to the options of the UI Builder, so the Dashboard follows the platform's component set.
- The risk score and category are produced by rules and by the evaluator; the application does not include a trained prediction model.
- The weekly and monthly digest schedules and the SMS dispatch described in the design brief are not yet configured; the Blueprint covers only the heart-attack screening and the approval process has a single CMO level.
- The Monthly Diabetic Review Reminder is inactive in the exported application, and alert e-mails are addressed to the logged-in user.
- The Patient profile has Modify and Delete rights on the Patient Master reports, and record-level criteria such as Patient_Email == zoho.loginuserid are not yet configured, so a patient could see more than their own data in the current permissions.
- Age is stored in a number field filled by a workflow rather than a formula field, and BMI is entered rather than derived from height and weight.
- The application time zone is set to America/Los_Angeles while the schedule runs in Asia/Kolkata.
- The application depends on Zoho Creator, so moving it to another platform would require rebuilding the forms and Deluge logic.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of HealthGuard AI is no-code configuration: forms, lookups, drop-downs, reports, kanban boards, the Dashboard, roles and profiles. Low-code Deluge was needed for the age calculation, the validation, the high-risk alert and the monthly reminder. Pro-code would be needed for the next level of capability: training and serving a machine-

learning risk model on real clinical data, integration with laboratory systems, and tamper-evident storage of clinical records. The point to move to pro-code is when a requirement depends on processing that Deluge or the platform components cannot express safely.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

HealthGuard AI is a Zoho Creator application for health risk screening and clinical decision support. It combines six linked forms, twelve reports, a Dashboard, form workflows for age calculation, validation and high-risk alerting, a schedule for diabetic review reminders, role-based access with a patient portal and a structured record for high-risk case approval with CMO review status.

The project shows that a structured screening and escalation workflow can be built on a low-code platform by one developer within a semester, with Deluge used only where configuration was not enough. The risk outputs are decision-support values that remain under the control of the physician and the CMO, and the report is explicit that the application does not diagnose or treat.

9.2 Future Enhancements

- Extend the Blueprint to all four disease modules (Patient Details, Details Analysis, Diseases Prediction and Risk Factor stages) and make the approval process multi-level with mandatory rejection notes for Critical cases.

- Activate the monthly diabetic reminder, add the daily vitals reminder, the weekly risk-trend digest and the monthly executive summary schedules, and address e-mails to the assigned staff and department heads.
- Restrict the Patient profile to view-only on Patient Master and add record-level criteria so that portal users see only their own records.
- Replace the number fields for age and BMI with formula fields, and set the application time zone to Asia/Kolkata.
- Add a machine-learning risk model hosted in Zoho Catalyst and called from Deluge, with physician review of every prediction.
- Add SMS dispatch to emergency care and integration with a hospital information system or laboratory system.

9.3 Learning Outcomes and Career Relevance

Building HealthGuard AI gave practical experience in data modelling with lookups, designing role-based access, writing Deluge validation and notification scripts, configuring form and schedule workflows, building a dashboard in the UI Builder and thinking about clinical safety and privacy in the design. These skills relate directly to roles such as low-code developer, solutions consultant and healthcare automation

engineer, and the experience of keeping a clinician in control of decision-support output is relevant to any role that introduces automation into a clinical process.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Space menu with one section per form	5.1	Figure 5.1
II	Forms with ≥ 8 distinct field types	Y	Name, Auto Number, Date, Date-Time, Number, Percentage, Phone, Email, Drop Down, Radio, Checkbox, Multi Line, Lookup	5.2	Figure 5.3
II	Lookup field (relationship between forms)	Y	Patient lookup in every assessment form	4.2, 5.2	Figure 4.2
II	Subform (parent–child data)	N	Not used; child records are separate forms	4.2	—
II	Field validations / layout rules	Y	Cancel Submit Invalid BP (systolic below 70)	5.5	Code 5.2
II	List report with filter, grouping, custom action	Partial	Six list reports; conditional formatting on oncology report; no custom actions	5.3	Figure 5.4
II	Kanban / Calendar / Timeline / Map report	Y	Six kanban reports (no calendar, timeline or map)	5.3	Figure 5.4
II	Pivot table / chart report	N	Not used; charts are on the Dashboard	5.3	—
II	Page / dashboard built with UI Builder	Y	Dashboard	5.4	Figure 5.5
II	Deluge – form events (validate / success / field actions)	Y	On validate, on user input, on load	5.5	Code 5.1–5.3
II	Deluge – functions, fetch, loops, collections	Partial	Fetch and for each loop used in scripts; no stand-alone functions	5.5	Appendix C
II	Email / in-app notifications	Y	High-risk alert; monthly diabetic reminder	5.5, 5.6	Code 5.3
II	Form workflow	Y	Three form workflows	5.6	Figure 5.6
II	Scheduled workflow	Partial	Monthly Diabetic Review Reminder (configured, inactive)	5.6	Figure 5.6
II	Approval process	Y	CMO High-Risk Case Approval on Clinical_Decision; approver CEO	5.6	Table 5.2
II	Blueprint	Y	HeartAttack Risk blueprint: 4 stages, 4 transitions	5.6, C.5	Figure C.1
II	Batch / report workflow	N	Not used	5.6	—

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Testing & debugging evidence	Partial	Test cases and defect log; actual results to be filled	6	Table 6.1
III	Integration with a Zoho service	Partial	E-mail through sendmail; no CRM or other Zoho app integration	5.7	Table 5.3
III	Connection / REST API call (invokeurl)	N	Not used	5.7	—
III	Webhook or Creator REST API	N	Not used	5.7	—
III	Users, roles and permission profiles	Y	Roles CEO and Staff; Cardiologist, Radiology, Patient profiles	5.8	Figure 5.7
III	Record-level access criteria / customer portal	Partial	Patient portal profile configured; no record-level criteria	5.8, 5.11	Figure 5.7
III	Zia / AI feature	N	Rule-based classification; no Zia or ML model	5.9	—
III	Analytics (charts / Zoho Analytics)	Y	Aggregates and three charts on the Dashboard	5.4, 5.9	Figure 5.5
IV	Performance optimisation applied & measured	Partial	Applied; not measured with timings	7.2	Table 7.1
IV	Deployment across environments	Y	Development to Production publishing	7.1	Figure 7.1
IV	Security controls (encryption, OAuth, permissions)	Partial	Profiles, portal, PII/ePHI flags; no OAuth connection used	7.3	Table 7.2
IV	Governance (audit trail, administration)	Partial	Clinical decision record and CMO status; no audit-trail form	7.4	—
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not used; justified in 5.10	5.10	—
V	Low-code limitations & spectrum analysis	Y	Sections 8.4 and 8.5	8.4, 8.5	—
—	Mobile app / portal access	Y	Patient portal profile; Creator mobile app	5.11	Figure 7.1

Total features fully implemented: 16 / 32 (9 partially implemented).

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	Good	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	Excellent	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	Satisfactory	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	Good	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	Satisfactory	

APPENDIX C – DELUGE SOURCE CODE

C.1 Auto-calculate Patient Age

Code C.1: Age calculation

```
// Script ID : DS-02 | Location : Form workflow (Patient_Master, on load)
// Purpose   : Auto-calculate age from date of birth
if(input.date_of_birth != null && !input.date_of_birth.isNull())
{
  currentDate = zoho.currentdate;
  dob = input.date_of_birth;
  ageYears = currentDate.getYear().toDecimal() - dob.getYear().toDecimal();
  currentMonth = currentDate.getMonth();
  currentDay = currentDate.getDay();
  dobMonth = dob.getMonth();
  dobDay = dob.getDay();
  if(currentMonth < dobMonth)
  {
    ageYears = ageYears - 1;
  }
  else if(currentMonth == dobMonth && currentDay < dobDay)
  {
    ageYears = ageYears - 1;
  }
  input.age = ageYears;
}
```

C.2 Cancel Submit Invalid BP

Code C.2: Blood pressure validation

```
// Script ID : DS-03 | Location : Form workflow (Cardiovascular_Assessment, on
validate)
// Purpose   : Reject an impossible systolic blood pressure
if(input.systolic_bp < 70)
{
  alert "Error: Systolic blood pressure cannot be below 70 mmHg. Please enter a
valid value.";
  cancel submit;
}
```

C.3 High Risk Cardio Alert

Code C.3: High-risk alert e-mail

```
// Script ID : DS-04 | Location : Form workflow (Cardiovascular_Assessment,
//              on user input of risk_score)
// Purpose   : E-mail the cardiologist when the risk score is above 80 percent
if(input.risk_score > 80)
{
  patientRecord = patient_master[ID == input.patient];
  cardiologistEmail = zoho.loginuserid;
  patientName = patientRecord.full_name.first_name.concat(" ")
                .concat(patientRecord.full_name.last_name);
  riskScoreValue = input.risk_score.toString();
  notificationMessage = "<p>Dear Cardiologist,</p>";
```

```
  notificationMessage = notificationMessage.concat("<p>A new cardiovascular
    assessment has been completed with a <strong>high risk score of "
    + riskScoreValue + "%</strong>.</p>");
  notificationMessage =
notificationMessage.concat("<p><strong>Patient:</strong> "
    + patientName + "</p>");
  notificationMessage = notificationMessage.concat("<p>Please review the
    assessment and take necessary action.</p>");
  sendmail
  [
    from :zoho.loginuserid
    to :cardiologistEmail
    subject :"High Risk Cardiovascular Assessment - Immediate Review Required"
    message :notificationMessage
    content type :HTML
  ]
}
```

C.4 Monthly Diabetic Review Reminder

Code C.4: Monthly reminder schedule

```
// Script ID : DS-05 | Location : Schedule (Metabolic_Assessment, Next Review
Date,
//              10:00 Asia/Kolkata)
// Purpose   : Remind the care team of diabetic reviews due in 30 days
upcomingReviews = metabolic_assessment[next_review_date in next 30 days];
if(upcomingReviews.count() > 0)
{
  emailBody = "<p>Dear Care Team,</p>";
  emailBody = emailBody.concat("<p>The following patients have upcoming
    diabetes review dates:</p><ul>");
  for each  assessment in upcomingReviews
  {
    patientRecord = patient_master[ID == assessment.patient.ID];
    patientName = "";
    if(!patientRecord.full_name.first_name.isNull())
    {
      patientName = patientRecord.full_name.first_name.toString();
      if(!patientRecord.full_name.last_name.isNull())
      {
        patientName = patientName.concat(" ")
                      .concat(patientRecord.full_name.last_name.toString());
      }
    }
    reviewDate = assessment.next_review_date.getDateTime();
    riskLevel = assessment.progression_stage.toString();
    emailBody = emailBody.concat("<li>" + patientName + " - Review Date: "
      + reviewDate.toString() + " - Risk Level: " + riskLevel + "</li>");
  }
  emailBody = emailBody.concat("</ul><p>Please ensure all necessary
    preparations are made for these reviews.</p>");
  sendmail
  [
    from :zoho.loginuserid
    to :zoho.loginuserid
    subject :"Upcoming Diabetes Review Dates"
```

```
    message :emailBody
    content type :HTML
  ]
}
```

C.5 HeartAttack Risk

Code C.5: HeartAttack Risk blueprint script

```
// Script ID : DS-01 | Location : Blueprint "HeartAttack Risk"
//   (Cardiovascular_Assessment, on form submission)
// Purpose : Complete the screening stages and time-stamp the risk output
// Transition "Final" (Risk Factor -> End), after-action script
if(input.prediction_timestamp == null)
{
    input.prediction_timestamp = zoho.currenttime;
}
if(input.risk_score > 80)
{
    // the e-mail itself is sent by the High Risk Cardio Alert workflow
    info "High risk score recorded for assessment " + input.ID;
}
```

Figure C.1 shows the blueprint canvas: the stages Patient Details, Details Analysis, Diseases Prediction and Risk Factor are joined by the transitions Analysis, Prediction, Risk and Final, between Start and End.

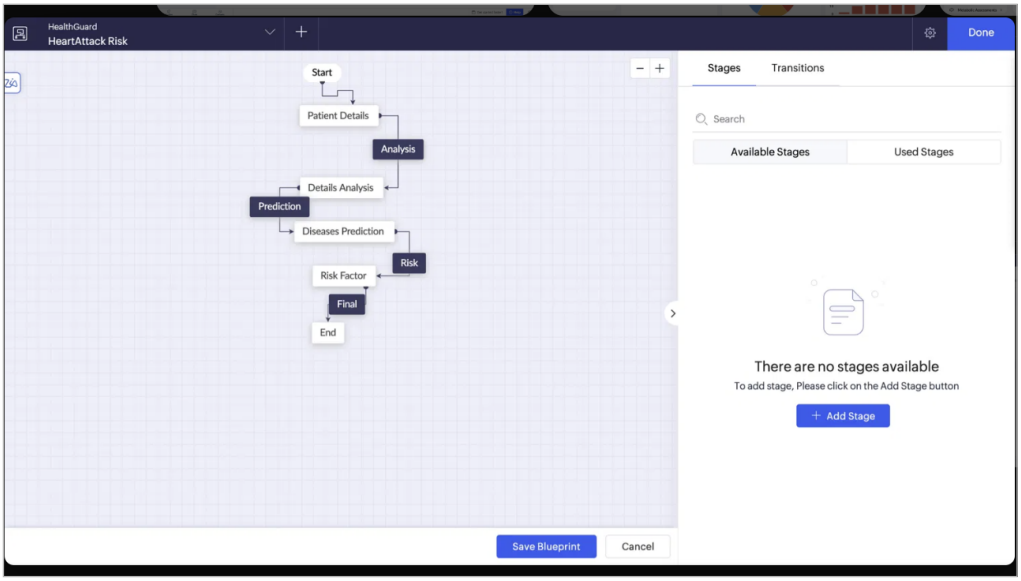


Figure C.1: HeartAttack Risk Blueprint in the Creator Blueprint builder

APPENDIX D – USER MANUAL

Open the HealthGuard AI application URL in a browser and sign in with your Zoho account. Portal users use the portal link and the credentials created for them. On a phone, install the Zoho Creator mobile app, sign in and open HealthGuard AI. The tabs you see depend on your permission profile.

URL : https://creatorapp.zoho.in/danylenoch.2024.cse_rajal670/clinical-risk-assessment-system#Page:DashBoard

CEO / Hospital Director. Open the Dashboard to see the average risk scores, the number of oncology screenings, the total high-risk cases and the CMO Review Status counts. Open All Clinical Decision Highrisk Case Approvals, select a case, review the notes and the physician's signature, and set the CMO Review Status to Approved, Override, Further Diagnostics Requested or Pending. Use the Disease Categories board to see the cases by disease. Manage users, roles and profiles in the Users & Control area.

Staff (Triage Nurse). Open Patient Master and click Add to register a patient with the full name, date of birth, gender, contact details, blood group and emergency contact; the age is calculated automatically. Use the Quick Links on the Dashboard to open the Cardiovascular Assessment, Oncology Screening or Clinical Decision forms. In the Cardiovascular Assessment, select the patient and enter the vitals; a systolic pressure below 70 mmHg is rejected.

Cardiologist. Open All Cardiovascular Assessments, review the vitals, ECG rhythm, chest pain type and troponin, and enter the risk score, category and recommendation. Scores above 80 percent send an alert e-mail. Create a Clinical Decision record for High or Critical cases and add the physician signature and the treatment protocol.

Radiology Staff. Open Oncology Screening and click Add; select the patient, the target organ, tobacco and alcohol exposure, tumour markers, weight loss, imaging findings and genetic markers, and set the Flag for Tumor Board Review when needed.

Patient (Portal). Sign in to the portal and open the reports to view your assessment summaries in read-only mode.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	Project Title	Register No.	Date
DANYL ENOCH	HealthGuard AI: Health Risk Screening System	240701092	10/11/2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	Register No.	Department / Section	Date of Viva
DANYL ENOCH	240701092	Computer Science and Engineering	10/11/2026

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners’ Remarks:

_____

INTERNAL EXAMINER EXTERNAL EXAMINER

Name: _____ Name: _____